UC SAN DIEGO / ECE 251C

8
Wavelet processing and perfect-reconstruction filter banks

Discrete wavelet transforms, synthesis, and alias cancellation

How can two downsampled channels retain every input sample?

8.1 From a wavelet representation to a processing method

The previous lecture turned orthogonal projections into filtering and downsampling. We now complete the other half of the construction: recover the input coefficients from the two output channels. We can then modify the wavelet coefficients between analysis and synthesis, giving a complete signal-processing method.

Recall the continuous-domain representation. Let ϕ and ψ generate an orthonormal wavelet system, with

$$\phi_{i,n}(t) = 2^{i/2}\phi(2^i t - n), \qquad \psi_{i,n}(t) = 2^{i/2}\psi(2^i t - n).$$

The approximation and detail coefficients are

$$a_i[n] = \langle f, \phi_{i,n}\rangle_{L^2}, \qquad d_i[n] = \langle f, \psi_{i,n}\rangle_{L^2}.$$

Keeping resolution zero as a reference gives

$$f(t) = \sum_{n\in\mathbb{Z}} a_0[n]\phi_{0,n}(t) + \sum_{i=0}^{\infty}\sum_{n\in\mathbb{Z}} d_i[n]\psi_{i,n}(t). \tag{8.1}$$

This is an equality in $L^2(\mathbb{R})$. The first term supplies the coarse approximation; each later level adds detail at a finer resolution.

The two-scale coefficients are

$$g_0[n] = \langle \phi, \phi_{1,n}\rangle_{L^2}, \qquad g_1[n] = \langle \psi, \phi_{1,n}\rangle_{L^2}. \tag{8.2}$$

These are discrete filters. We use real-valued ϕ, ψ and real filters below, as in the Haar example. The inner product still conjugates its second argument; that conjugation has no effect on these real functions.

8.2 The general wavelet analysis pipeline

A computer does not receive the whole function $f \in L^2(\mathbb{R})$. It receives a sequence from an acquisition device: pixel measurements, audio measurements, or another collection of samples. Denote that sequence by $a_I[n]$, where I is the finest available resolution. We fix scaling and wavelet functions ϕ, ψ from an orthonormal MRA.

The fundamental modeling assumption is

$$\boxed{f(t) \approx f_I(t) := \sum_{n\in\mathbb{Z}} a_I[n]\phi_{I,n}(t).} \tag{8.3}$$

In the ideal coefficient model, $a_I[n] = \langle f, \phi_{I,n}\rangle_{L^2}$ and $f_I = P_I f$. For a real device, interpreting its measurements as these coefficients is an assumption about the acquisition process. Without coefficient processing, perfect reconstruction recovers a_I; it does not remove the approximation of f by f_I.

Choose an integer $K \geq 1$ for the number of decomposition levels. The complete procedure has four steps. Bars will mark processed coefficients and signals.

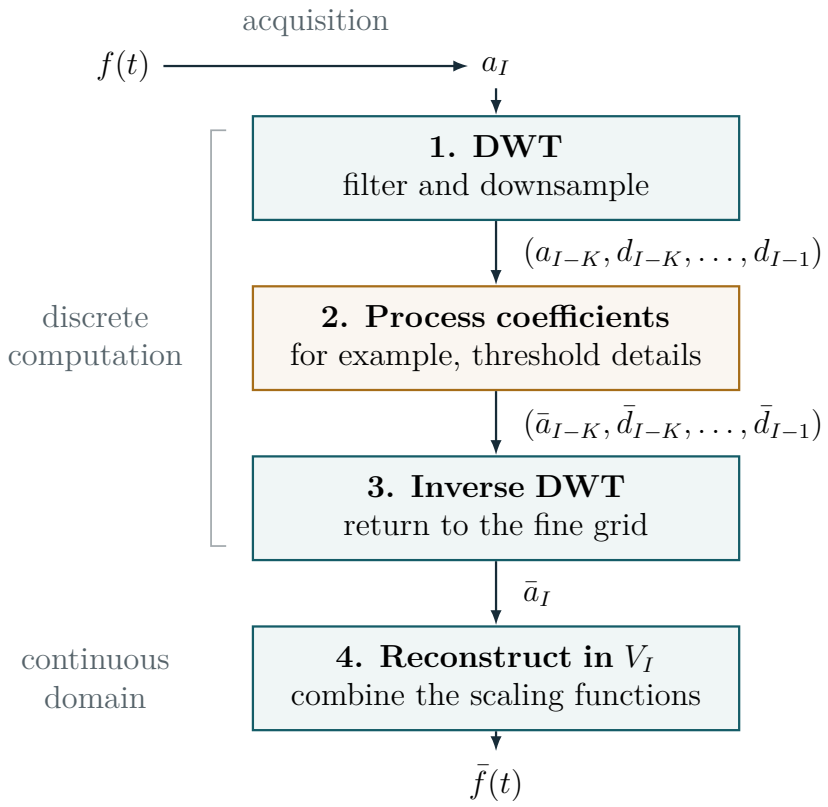


Figure 8.1: The complete pipeline. Acquisition supplies a_I; the DWT reorganizes it by scale, processing changes the coefficients, and the inverse DWT produces $\bar{a}_I$. The last step represents that output as a function $\bar{f} \in V_I$.

Step 1: perform a K-level DWT. At each level, apply the low-pass and high-pass analysis filters and downsample both outputs by two. Continue with only the approximation output, retaining each detail sequence at its own level:

$$a_I \xrightarrow{\text{DWT}} (a_{I-K}, d_{I-K}, \ldots, d_{I-1}).$$

Thus there is one coarse approximation and K detail sequences. For example, three levels give $(a_{I-3}, d_{I-3}, d_{I-2}, d_{I-1})$, not just the coarsest approximation a_{I-3}.

Step 2: process the wavelet coefficients. Modify this collection to obtain

$$(\bar{a}_{I-K}, \bar{d}_{I-K}, \ldots, \bar{d}_{I-1}).$$

A common choice is to leave a_{I-K} unchanged and threshold only the details. More generally, the processing rule can act on the entire collection. A bar indicates the output of this processing; it does not denote complex conjugation, for which we continue to use $*$. Hats remain reserved for Fourier transforms.

Step 3: perform the inverse DWT. Starting from the processed coarse approximation, reintroduce the details from coarsest to finest:

$$(\bar{a}_{I-K}, \bar{d}_{I-K}, \ldots, \bar{d}_{I-1}) \xrightarrow{\text{IDWT}} \bar{a}_I.$$

The first inverse stage combines $\bar{a}_{I-K}$ and $\bar{d}_{I-K}$ to form $\bar{a}_{I-K+1}$. The next stage uses $\bar{d}_{I-K+1}$, and so on until resolution I is reached. This reconstructs the modified coordinates; it does not undo the processing.

Step 4: form the processed continuous-domain signal. Use the same fine-resolution scaling functions as in the acquisition model:

$$\boxed{\bar{f}(t) \coloneqq \sum_{n \in \mathbb{Z}} \bar{a}_I[n] \phi_{I,n}(t).} \tag{8.4}$$

The inverse DWT returns a sequence; this expansion identifies the function that the sequence represents. Both f_I and $\bar{f}$ lie in V_I, with possibly different coefficients.

Without coefficient modification. An exact inverse DWT returns $\bar{a}_I = a_I$, so $\bar{f} = f_I$. Equality with the underlying f additionally requires the acquisition model to be exact. The rest of the lecture explains how filtering and rate changes implement these forward and inverse operations.

8.3 The multilevel discrete wavelet transform

At one level, the analysis filters are the reversed two-scale filters:

$$h_0[n] = g_0[-n], \qquad h_1[n] = g_1[-n].$$

For complex filters the corresponding formula is $h_\ell[n] = g_\ell^*[-n]$; the real-valued setting reduces it to reversal. Writing $\mathrm{H}_\ell x = h_\ell * x$, the two outputs are

$$\boxed{a_i[m] = (h_0 * a_{i+1})[2m], \qquad d_i[m] = (h_1 * a_{i+1})[2m].} \tag{8.5}$$

The symbol $\downarrow 2$ means *retain the even-indexed entries*: $(\downarrow 2\, v)[m] = v[2m]$. Filtering combines neighboring coefficients; downsampling then expresses each output on the coarser grid.

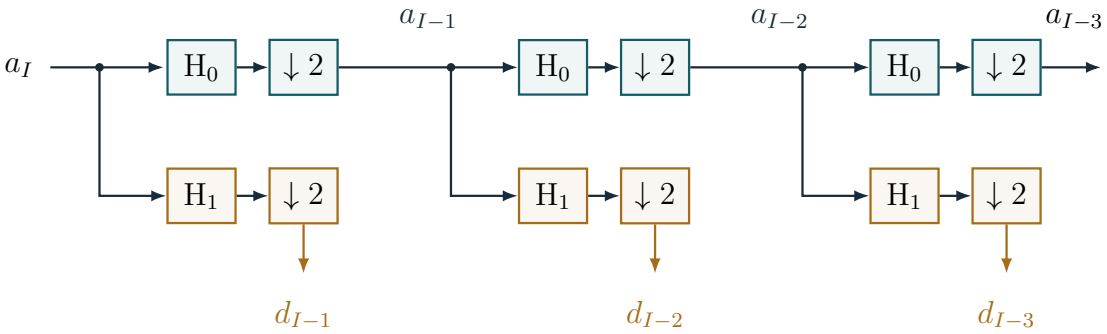


Figure 8.2: A three-level DWT. Only the approximation output is split again. Each detail output is retained at the resolution where it was computed.

The first split replaces a_I by (a_{I-1}, d_{I-1}). The second replaces only a_{I-1}, and the third replaces only a_{I-2}. Thus the final stored representation is

$$a_{I-3}, \quad d_{I-3}, \quad d_{I-2}, \quad d_{I-1}.$$

These sequences together describe the same f_I:

$$f_I = \sum_n a_{I-3}[n]\phi_{I-3,n} + \sum_{i=I-3}^{I-1} \sum_n d_i[n]\psi_{i,n}. \tag{8.6}$$

The finest detail index is $I-1$, not I. For K decomposition levels, the coarsest retained approximation is a_{I-K}.

The algorithm uses the filters, not evaluations of ϕ or ψ. The functions explain the coordinates; the filters specify the computation.

8.4 Processing changes the coefficients, not the basis

A change of coordinates alone does not remove noise or compress a signal. It preserves the information in a_I. The purpose of the wavelet representation is to expose coefficients that can be modified in a useful way. A typical choice is to keep the coarse approximation and threshold the detail coefficients.

Example: hard thresholding. For a threshold $\lambda > 0$, define

$$\mathcal{T}_\lambda(c) = \begin{cases} 0, & |c| < \lambda, \\ c, & |c| \geq \lambda. \end{cases}$$

Small coefficients are set to zero; the others are retained. With K decomposition levels, the processed coefficients are

$$\bar{a}_{I-K} = a_{I-K}, \qquad \bar{d}_i[n] = \mathcal{T}_\lambda(d_i[n]), \quad I-K \leq i < I.$$

For three levels, this acts on $d_{I-3}, d_{I-2}, d_{I-1}$. Choosing the threshold is a separate design question; here we are identifying where the processing enters.

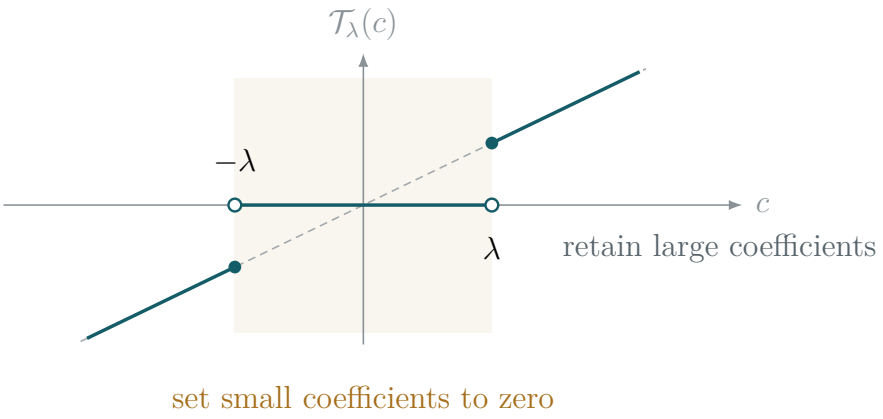


Figure 8.3: Hard thresholding for a real coefficient. The dashed line is the identity map. The jump at each threshold is part of the processing rule.

Thresholding is nonlinear. For example, each of two coefficients equal to $3\lambda/4$ is sent to zero, but their sum $3\lambda/2$ is retained. Thus thresholding a sum is not the same as adding the thresholded values.

Think analog, act digital

The MRA interprets the signal and its scales; the computation uses only sequences and filters. After processing, the inverse DWT faithfully synthesizes the *modified* coordinates. It does not undo thresholding. The resulting $\bar{a}_I$ specifies $\bar{f}$ through equation (8.4).

8.5 Recovering the fine coefficients

To invert one stage, we must combine the coarse approximation and its detail:

$$f_{i+1} = \sum_m a_i[m]\phi_{i,m} + \sum_m d_i[m]\psi_{i,m}.$$

The two-scale equations express both families using the finer scaling functions. With the same masks as in equation (8.2),

$$\phi_{i,m} = \sum_n g_0[n-2m]\phi_{i+1,n},$$
$$\psi_{i,m} = \sum_n g_1[n-2m]\phi_{i+1,n}.$$

For example, substitute $2^i t - m$ into $\phi(t) = \sum_k g_0[k]\sqrt{2}\,\phi(2t-k)$ and multiply by $2^{i/2}$. The fine index is $n = 2m + k$, so the mask index is $k = n - 2m$.

Substituting both expansions and collecting the coefficient of $\phi_{i+1,n}$ gives

$$\begin{aligned} f_{i+1} &= \sum_m \sum_n a_i[m]g_0[n-2m]\phi_{i+1,n} \\ &\quad + \sum_m \sum_n d_i[m]g_1[n-2m]\phi_{i+1,n} \\ &= \sum_n \left(\sum_m a_i[m]g_0[n-2m] + \sum_m d_i[m]g_1[n-2m] \right) \phi_{i+1,n}. \end{aligned}$$

Uniqueness of the coordinates in this orthonormal basis yields

$$\boxed{a_{i+1}[n] = \sum_m a_i[m]g_0[n-2m] + \sum_m d_i[m]g_1[n-2m].} \tag{8.7}$$

Why upsampling appears. Define zero-insertion upsampling by

$$(\uparrow 2\,c)[n] := \begin{cases} c[n/2], & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases} \tag{8.8}$$

Then convolution with g satisfies

$$(g * (\uparrow 2\,c))[n] = \sum_k g[n-k](\uparrow 2\,c)[k] = \sum_m g[n-2m]c[m].$$

Only even $k = 2m$ survive. Consequently equation (8.7) is exactly

$$a_{i+1} = g_0 * (\uparrow 2\,a_i) + g_1 * (\uparrow 2\,d_i). \tag{8.9}$$

A synthesis bank. The inverse stage first inserts zeros, then filters each channel, and finally adds the results. Upsampling places each coarse coefficient on the fine grid; the filters then distribute its contribution among the finer coefficients.

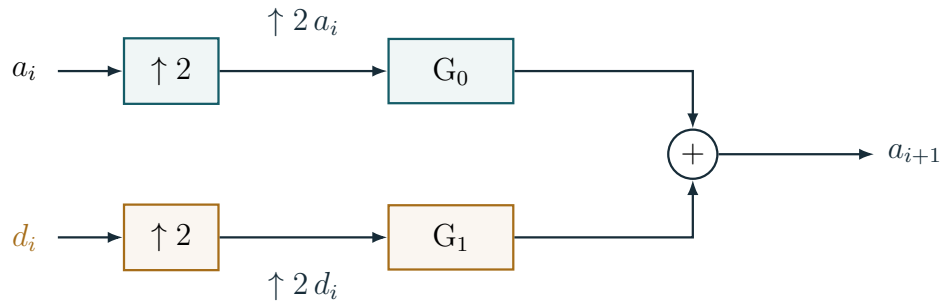


Figure 8.4: Synthesis reverses one analysis stage. Here $\mathrm{G}_\ell c = g_\ell * c$. The order is *upsample, then filter*, as required by the index $n - 2m$ in the synthesis formula.

Zero insertion is not interpolation. Consider the three nonzero entries $c[0] = 1$, $c[1] = 2$, $c[2] = -1$. Upsampling changes their locations to $0, 2, 4$ and fills the odd locations with zeros. It does not infer missing values.

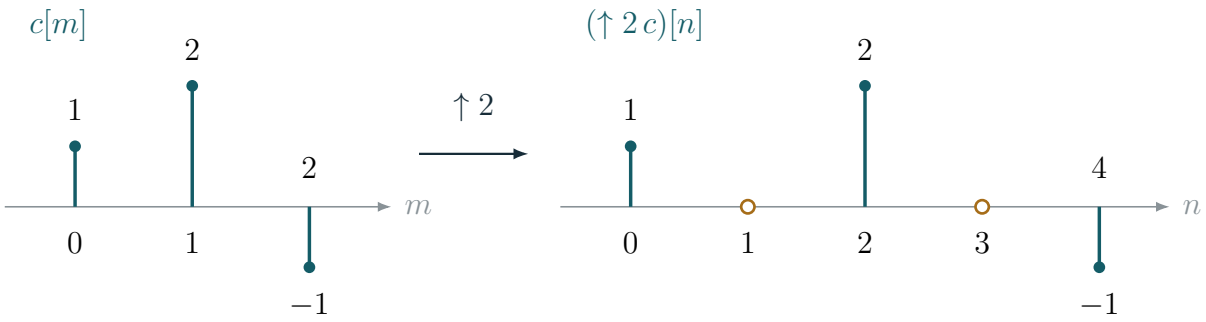


Figure 8.5: The open gold circles mark inserted zeros. Filtering after this step supplies the contributions needed on the fine grid.

The order of these operations matters. Filtering c on the coarse grid and then inserting zeros would, in general, give a different result with the same filter coefficients.

For processed coefficients, the same synthesis formula applies with bars on every a_i and d_i. A three-level inverse DWT first combines $\bar{a}_{I-3}$ and $\bar{d}_{I-3}$ to produce $\bar{a}_{I-2}$, then uses $\bar{d}_{I-2}$, and finally $\bar{d}_{I-1}$. The result is the processed sequence $\bar{a}_I$.

8.6 Downsampling followed by upsampling

Before analyzing a complete filter bank, isolate its rate changes. Let

$$c[m] = v[2m], \qquad w = \uparrow 2\, c.$$

Then $w[2m] = v[2m]$ and $w[2m+1] = 0$. The cascade returns the sequence to its original index grid, but it does not restore the discarded odd entries.

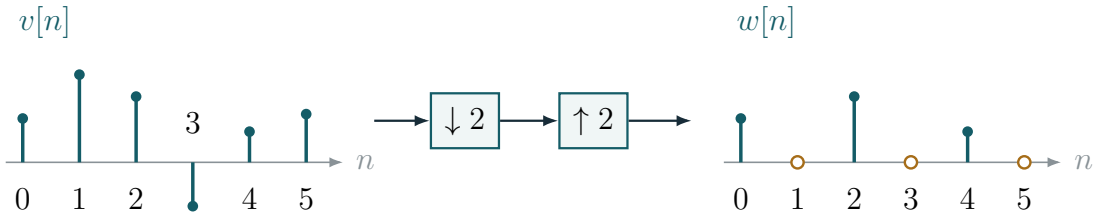


Figure 8.6: Only the even entries survive. The two rate changes together act as an alternating keep/erase mask on the original grid.

The even-index mask has a useful algebraic expression:

$$\frac{1+(-1)^n}{2} = \begin{cases} 1, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

Therefore

$$\boxed{w[n] = \frac{1}{2}\big(v[n] + (-1)^n v[n]\big).} \tag{8.10}$$

To transform the second term, recall the bilateral z-transform $V(z) = \sum_n v[n] z^{-n}$. Since $(-1)^{-n} = (-1)^n$ for integers n,

$$\sum_n (-1)^n v[n] z^{-n} = \sum_n v[n](-z)^{-n} = V(-z).$$

Thus the combined rate changes give

$$\boxed{W(z) = \frac{1}{2}\big(V(z) + V(-z)\big).} \tag{8.11}$$

On the unit circle, $-\mathrm{e}^{\mathrm{j}\omega} = \mathrm{e}^{\mathrm{j}(\omega+\pi)}$. The term $V(-z)$ is a frequency-shifted copy, displaced by π. This is the alias term that must cancel between the two channels.

8.7 The two-channel reconstruction formula

We can now examine the analysis and synthesis banks together. The input x is an arbitrary discrete sequence; in the wavelet algorithm it is one of the approximation sequences a_{i+1}. There is no coefficient modification in the following diagram.

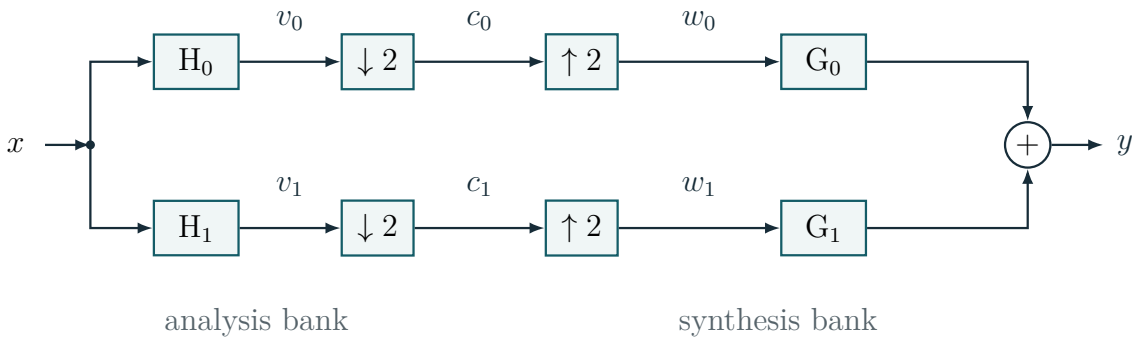


Figure 8.7: Each analysis branch filters and downsamples. Each synthesis branch upsamples and filters. Both branch outputs are added on the original index grid.

For the algebra below, take the filters to be FIR and begin with a finite input, so every transform is a finite sum of powers of z. In branch $\ell \in \{0, 1\}$,

$$V_\ell(z) = H_\ell(z)X(z).$$

Applying equation (8.11), then the synthesis filter, gives the branch contribution

$$G_\ell(z)W_\ell(z) = \frac{1}{2}G_\ell(z)\big(H_\ell(z)X(z) + H_\ell(-z)X(-z)\big).$$

Adding the two branches and grouping the two input terms yields

$$\begin{aligned} Y(z) &= \frac{1}{2}G_0(z)\big(H_0(z)X(z) + H_0(-z)X(-z)\big) \\ &\quad + \frac{1}{2}G_1(z)\big(H_1(z)X(z) + H_1(-z)X(-z)\big) \\ &= \underbrace{\frac{G_0(z)H_0(z) + G_1(z)H_1(z)}{2}}_{T(z)} X(z) \\ &\quad + \underbrace{\frac{G_0(z)H_0(-z) + G_1(z)H_1(-z)}{2}}_{A(z)} X(-z). \end{aligned} \tag{8.12}$$

The term $T(z)$ describes the gain and phase applied to the original spectrum. The term $A(z)$ multiplies the shifted spectrum. The system is not described by a single convolution response unless this alias contribution disappears.

8.8 Two conditions for perfect reconstruction

Definition. The filter bank has *perfect reconstruction* (PR) if, without coefficient processing,

$$y[n] = x[n - L] \qquad \text{for every input } x,$$

for a fixed integer L. A nonnegative L is a delay; $L = 0$ gives exact sample-by-sample reconstruction. Its transform is $Y(z) = z^{-L}X(z)$: all frequencies must have unit gain and the same time delay.

Proposition 8.1 (Vetterli, 1986)**.** For the two-channel FIR bank above, perfect reconstruction with shift L is equivalent to

$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2z^{-L}, \qquad \text{no distortion,} \tag{8.13}$$

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0, \qquad \text{no aliasing.} \tag{8.14}$$

The factor of two compensates for the factor $1/2$ introduced by the even-index selector.

Proof. If both identities hold, equation (8.12) reduces to $Y(z) = z^{-L}X(z)$.

For necessity, the reconstruction rule must hold for every input. With $x[n] = \delta[n]$, we have $X(z) = X(-z) = 1$, so

$$T(z) + A(z) = z^{-L}.$$

With $x[n] = \delta[n - 1]$, we have $X(z) = z^{-1}$ and $X(-z) = -z^{-1}$. Hence

$$T(z) - A(z) = z^{-L}.$$

Adding and subtracting gives $T(z) = z^{-L}$ and $A(z) = 0$, which are exactly the two stated identities. □

Why alias cancellation is possible. Each downsampled branch can contain aliasing. Perfect reconstruction does not require each branch to reconstruct the input separately. It requires their *sum* to cancel the alias terms while retaining the desired signal.

Why a delay is allowed. A delay changes only the time origin. It does not alter relative amplitudes or waveform shape. A frequency-dependent gain, by contrast, would change the signal even after the output is shifted back.

8.9 The Haar analysis and synthesis filters

For Haar, the synthesis filters have their nonzero taps at 0 and 1:

$$g_0[n] = \frac{\delta[n] + \delta[n-1]}{\sqrt{2}}, \qquad g_1[n] = \frac{\delta[n] - \delta[n-1]}{\sqrt{2}}.$$

Reversing these sequences places the analysis taps at -1 and 0:

$$h_0[n] = \frac{\delta[n+1] + \delta[n]}{\sqrt{2}}, \qquad h_1[n] = \frac{\delta[n] - \delta[n+1]}{\sqrt{2}}.$$

The negative index is intentional: with this alignment, analysis combines $x[2m]$ and $x[2m+1]$.

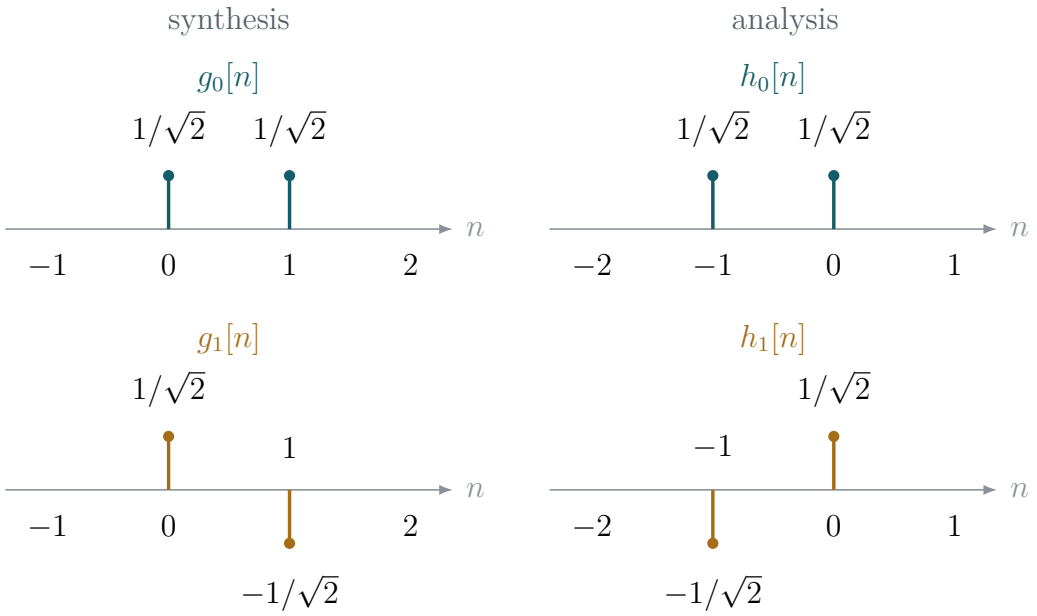


Figure 8.8: Reversal changes the tap locations, not their values. The high-pass filters have opposite signs at neighboring indices.

Using $\delta[n-k] \xleftrightarrow{\mathcal{Z}} z^{-k}$, we obtain

$$\begin{aligned} G_0(z) &= \frac{1 + z^{-1}}{\sqrt{2}}, \quad H_0(z) = \frac{1 + z}{\sqrt{2}}, \\ G_1(z) &= \frac{1 - z^{-1}}{\sqrt{2}}, \quad H_1(z) = \frac{1 - z}{\sqrt{2}}. \end{aligned} \tag{8.15}$$

In particular, the tap at $n = -1$ contributes z, not z^{-1}. We now substitute these four expressions into both PR identities.

No distortion. Expand each product before adding:

$$\begin{aligned} G_0(z)H_0(z) + G_1(z)H_1(z) &= \frac{1}{2}(1+z^{-1})(1+z) + \frac{1}{2}(1-z^{-1})(1-z) \\ &= \frac{1}{2}\big(2 + z + z^{-1}\big) + \frac{1}{2}\big(2 - z - z^{-1}\big) \\ &= 2. \end{aligned}$$

Thus equation (8.13) holds with $L = 0$. This zero-delay alignment uses the analysis tap at -1; it is not a claim that both analysis filters are causal.

No aliasing. First substitute $-z$ into the analysis responses:

$$H_0(-z) = \frac{1-z}{\sqrt{2}}, \qquad H_1(-z) = \frac{1+z}{\sqrt{2}}.$$

Then

$$\begin{aligned} G_0(z)H_0(-z) + G_1(z)H_1(-z) &= \frac{1}{2}(1+z^{-1})(1-z) + \frac{1}{2}(1-z^{-1})(1+z) \\ &= \frac{1}{2}\big(z^{-1} - z\big) + \frac{1}{2}\big(z - z^{-1}\big) \\ &= 0. \end{aligned}$$

The two alias contributions are nonzero in general, but are exact opposites. Combining both calculations in equation (8.12) gives $Y(z) = X(z)$.

Seeing the cancellation on the unit circle. Write

$$T_\ell(z) = \tfrac{1}{2}G_\ell(z)H_\ell(z), \qquad A_\ell(z) = \tfrac{1}{2}G_\ell(z)H_\ell(-z).$$

At $z = \mathrm{e}^{\mathrm{j}\omega}$, the identities $z + z^{-1} = 2\cos\omega$ and $z - z^{-1} = 2\mathrm{j}\sin\omega$ give

$$\begin{aligned} T_0 &= \cos^2(\omega/2), \quad T_1 = \sin^2(\omega/2), \\ A_0 &= -\tfrac{\mathrm{j}}{2}\sin\omega, \quad A_1 = \tfrac{\mathrm{j}}{2}\sin\omega. \end{aligned}$$

The first two responses add to one; the last two add to zero.

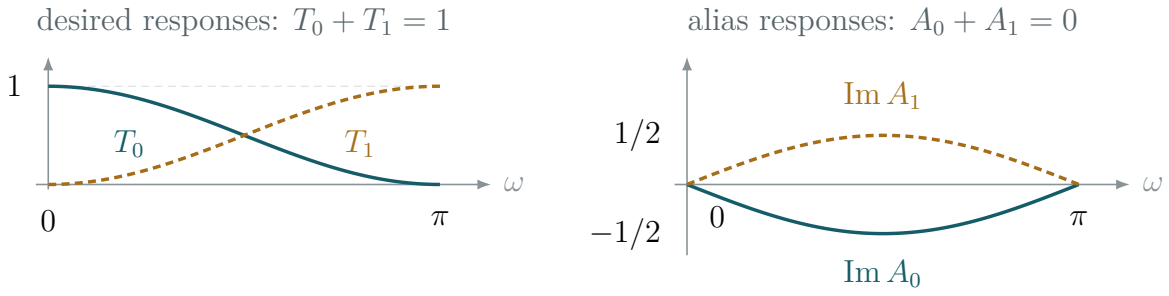


Figure 8.9: For the Haar bank, the desired parts complement one another while the imaginary alias parts cancel. The plots show branch contributions *after* synthesis, including their factor 1/2.

8.10 Perfect reconstruction on each pair of samples

A direct check uses $(h_\ell * x)[2m] = \sum_k h_\ell[k]x[2m-k]$. With only the taps at $k = -1, 0$ nonzero, analysis becomes

$$a[m] = \frac{x[2m] + x[2m+1]}{\sqrt{2}}, \qquad d[m] = \frac{x[2m] - x[2m+1]}{\sqrt{2}}. \tag{8.16}$$

Let $u = \uparrow 2\, a$ and $v = \uparrow 2\, d$. Synthesis is

$$y[n] = \frac{u[n] + u[n-1]}{\sqrt{2}} + \frac{v[n] - v[n-1]}{\sqrt{2}}.$$

For $n = 2m$, the entries $u[n-1]$ and $v[n-1]$ vanish. For $n = 2m+1$, the entries $u[n]$ and $v[n]$ vanish. Consequently

$$y[2m] = \frac{a[m] + d[m]}{\sqrt{2}} = \frac{2x[2m]}{2} = x[2m],$$
$$y[2m+1] = \frac{a[m] - d[m]}{\sqrt{2}} = \frac{2x[2m+1]}{2} = x[2m+1].$$

In the sum, the odd sample cancels; in the difference, the even sample cancels. Both entries are recovered. Retaining both channels, rather than either channel alone, preserves the pair.

The Haar bank is orthogonal. In matrix form,

$$\begin{pmatrix} a[m] \\ d[m] \end{pmatrix} = U \begin{pmatrix} x[2m] \\ x[2m+1] \end{pmatrix}, \qquad U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

A direct multiplication gives

$$U^{\mathsf{T}} U = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I_2.$$

The inverse is $U^{\mathsf{T}} = U$, and $|a[m]|^2 + |d[m]|^2 = |x[2m]|^2 + |x[2m+1]|^2$. Summing over m shows that the complete analysis bank preserves sequence energy:

$$\|a\|_{\ell^2}^2 + \|d\|_{\ell^2}^2 = \|x\|_{\ell^2}^2.$$

This is the discrete counterpart of the orthogonal decomposition $V_{i+1} = V_i \oplus W_i$.

> **What perfect reconstruction guarantees**
> Without coefficient processing, synthesis returns exactly the acquired sequence. With processing, it returns the signal specified by the modified coefficients. Neither assertion removes the approximation made during acquisition.